
Compositionality & the Geometry of CLIP

Davide Berasi

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Outline

- Introduction to compositionality
- Compositionality in deep learning
- Compositionality in CLIP
- Geometrical properties of CLIP embeddings

Understanding complex concepts

- Humans can understand new complex concepts never seen before.
- How can we do it?

*“A purple elephant fixing
a refrigerator.”*



Understanding complex concepts

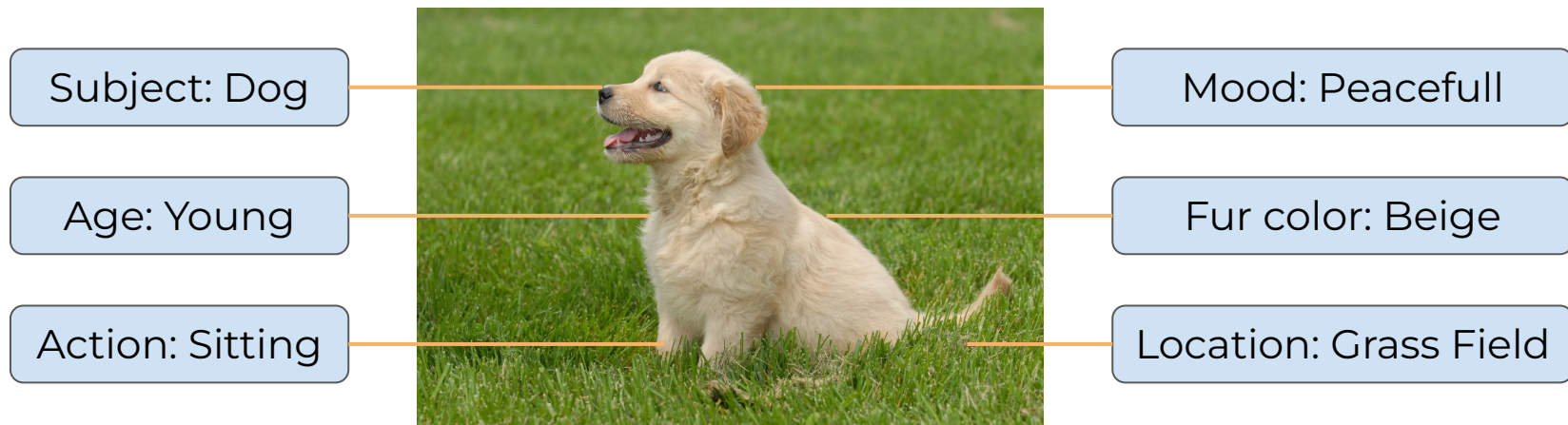
- Humans can understand new complex concepts never seen before.
- How can we do it? → **Compositional Understanding**

*“A purple elephant fixing
a refrigerator.”*



Compositionality

The meaning of an expression is a function of the meanings of its parts and of the way they are syntactically combined.^[1]



[1] Partee, Barbara, "Compositionality" (1984)

Compositional Understanding

Compositional Understanding is the ability to exploit compositionality.

Humans do Compositional Understanding:

- We decompose complex concepts into their primitive parts.
- We compose primitive meanings into complex concepts.

Compositional Understanding

We use this mechanism in:

- **Language:** a sentence combines words via grammatical rules,
- **Visual perception:** in a scene, we identify objects, their properties, relations, ...

It is at the core of intelligence: it is efficient and enables generalization!

Compositional Understanding

- **Efficiency:** a finite set of primitives and rules gives an exponential number of combinations.

(E.g., in english, we can express any thought with ~50k words.)

- **Generalization:** we can understand/imagine new concepts.



Compositional
Understanding
→



Compositionality in Deep Learning

Classical symbolic AI (1950-1980):

- Explicit symbols + rules (e.g., rule-based systems, logic programs).
- Compositionality was explicitly imposed.

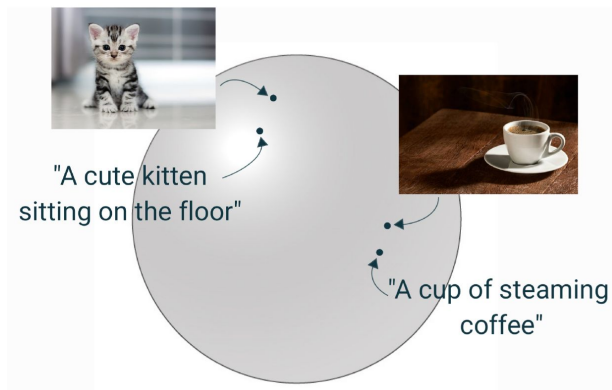
Modern deep learning:

- Directly learns from data. Compositionality **is not imposed** a priori.
- Does any compositional behaviour emerge?

Let's look at CLIP...

CLIP represents complex concepts (text & images) as vectors.

- Image encoder gives $\mathbf{u}_I \in \mathbb{R}^{512}$
- Text encoder gives $\mathbf{u}_T \in \mathbb{R}^{512}$
- Cosine similarity measures semantic similarity.



Are CLIP embeddings compositional?

Two possibilities:

1. CLIP is not compositional:

- It simply memorizes all the concepts in the training set.

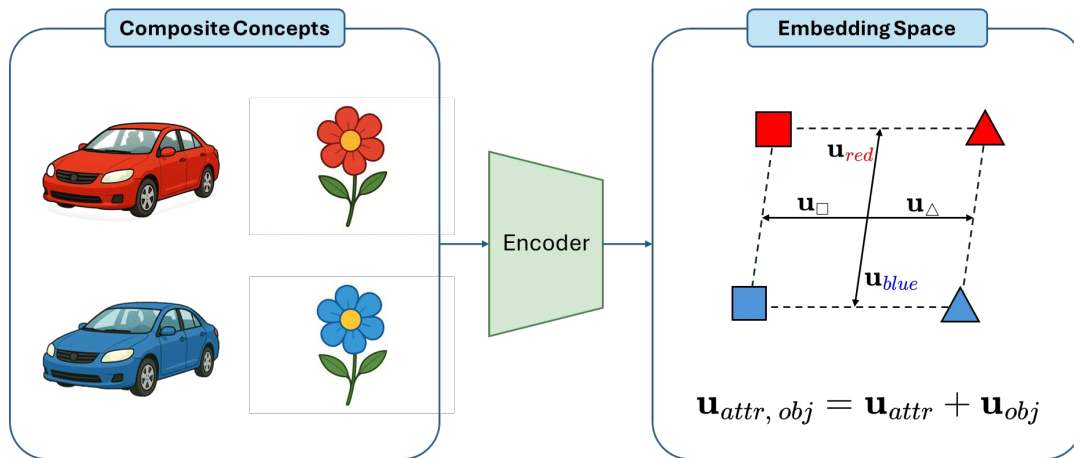
2. CLIP is compositional:

- It learns primitive concepts and how to combine them.
- Embeddings are organized in compositional structures.

CLIP compositionality

Let's consider a simple setting:

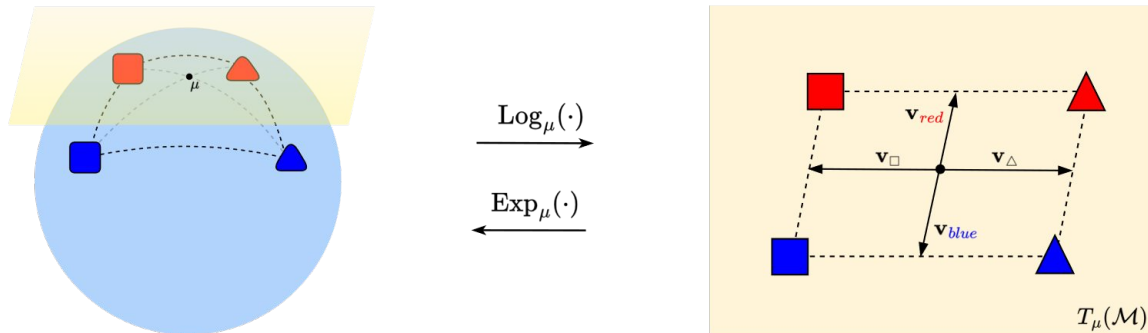
- Set of complex concepts $Z = \{a_1, \dots, a_N\} \times \{o_1, \dots, o_M\}$
- Embeddings are *decomposable* if $\mathbf{u}_{(\text{attr}, \text{obj})} = \mathbf{u}_{\text{attr}} + \mathbf{u}_{\text{obj}}$



CLIP compositionality

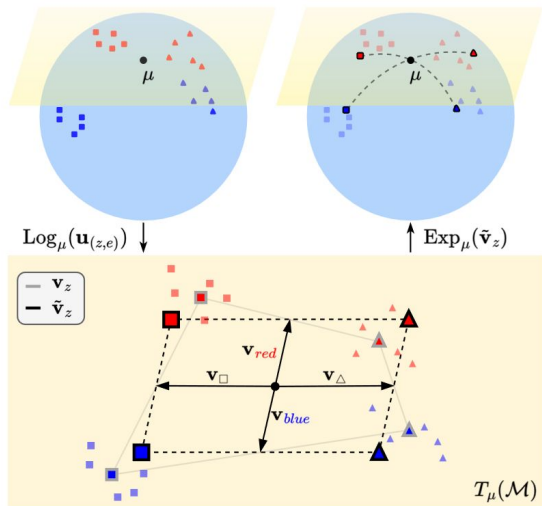
CLIP embeddings are actually on a sphere...let's account for that.

$$\mathbf{u}_{attr, obj} = \text{Exp}_\mu(\mathbf{v}_{attr} + \mathbf{v}_{obj})$$



CLIP compositionality

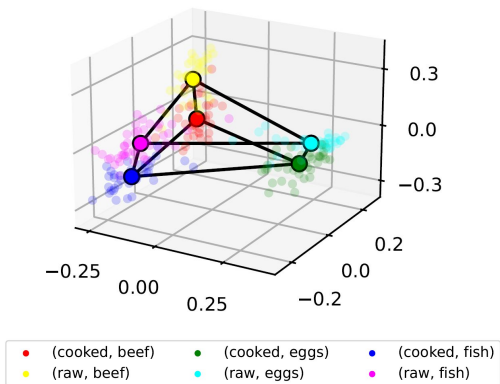
1. Embeddings are not perfectly decomposable.
2. We search the decomposable set that best approximates them.
3. Primitive directions of the ‘best decomposable approximation’ are vector means.
4. We show this is a good approximation of the original embeddings.



CLIP compositionality

Experiments show that the approximation is good, meaning CLIP embeddings are close to be decomposable.

Projection of 2x3 concepts set.

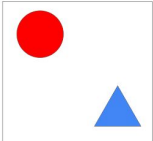


Inverting decomposable approximations.



Shortcomings in CLIP compositionality

- CLIP is bad at more complex compositions.
- It struggles with words order (treats text as a ‘bag-of-words’).
- Actually, CLIP may encode order information, but fail in cross-modal alignment^[Koishigarina et al., 2026].

		Cosine Similarity (CLIP)
<i>Image Prompt:</i> 	Attribute Binding	'red circle blue triangle' 0.3308 'red triangle blue circle' 0.3445 X
	Spatial Relationships	'circle above triangle' 0.2932 'circle below triangle' 0.2986 X
	Negation	'circle with triangle' 0.2925 'circle without triangle' 0.2927 X

Yuksekgonul et al., "When and why vision-language models behave like bags-of-words, and what to do about it?", ICLR23

Kamath et al., "What's 'up' with vision-language models? investigating their struggle with spatial reasoning", EMNLP23

Lewis et al., "Does CLIP Bind Concepts? Probing Compositionality in Large Image Models", EACL24

Kang et al., "Is CLIP ideal? No. Can we fix it? Yes!", ICCV25

Koishigarina et al., "CLIP Behaves like a Bag-of-Words Model Cross-modally but not Uni-modally", ICLR26

Latent space geometry

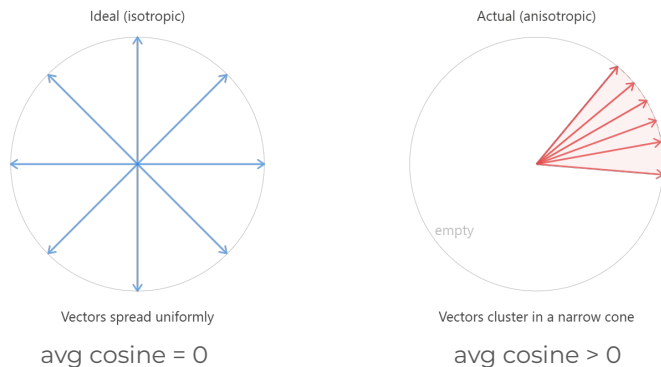
Besides compositional structures, the CLIP latent space has other interesting properties:

- Narrow Cone Effect
- Modality Gap
- Orthogonality and Superposition
- Spherical geometry: Log and Exp maps

The narrow cone effect

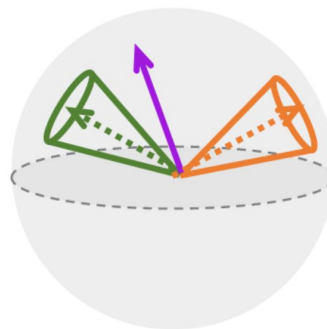
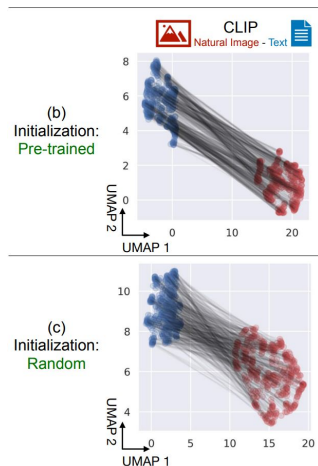
- Embeddings from deep networks cluster in a narrow cone.
- E.g., for CLIP image encoder, avg. cosine is ~ 0.5 (60°).

RK: In $\mathbb{R}^{512}$ a 60° cone occupies $\sim 10^{-153}\%$ of the hypersphere surface.



Modality Gap

- Text and image embeddings are in two narrow separate cones.
- Remember this if you combine embedding across modalities...



Orthogonality and Superposition

- Concepts can be grouped in *attributes* (e.g., subject, shape, color, size, ...)
- Varying a concept within an attribute should not change concepts within the other attributes. Indeed:
 - Concepts from different attributes are roughly orthogonal.
 - Concepts within the same attribute are not orthogonal.

Orthogonality and Superposition

How can we have thousands of orthogonal concepts in the ‘small’ CLIP latent space ($\mathbb{R}^{512}$)?

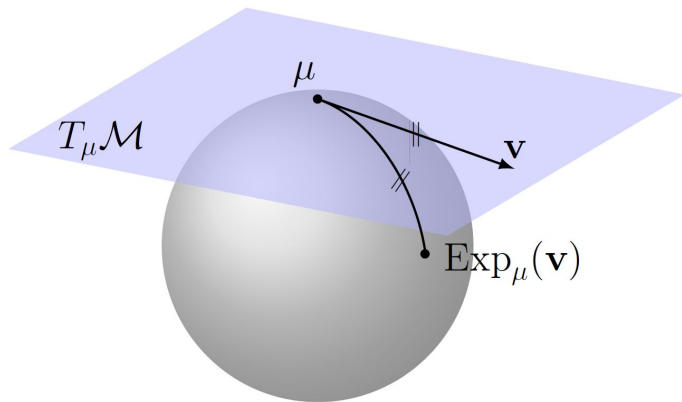
Superposition [from Johnson-Lindenstrauss lemma]:

In $\mathbb{R}^N$, you can fit $\sim \exp(\epsilon N)$ vectors with angles in $(90^\circ - \epsilon, 90^\circ + \epsilon)$.

Ex: in $\mathbb{R}^{100}$, we can fit 10k vectors with pairwise angles in $(89^\circ, 91^\circ)$.

Spherical geometry

- When playing with normalized embeddings (arithmetic, projections), the sphere geometry can lead to small distortions.
- A solution is using tangent plane approximation (*log* and *exp* maps).



The exponential and logarithmic maps can be expressed in closed form on the unit sphere $\mathbb{S}^{d-1}$. For any point of tangency $\mu \in \mathbb{S}^{d-1}$, we have

$$\text{Exp}_\mu(\mathbf{v}) = \cos(\|\mathbf{v}\|)\mu + \sin(\|\mathbf{v}\|)\frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \mathbf{v} \in T_\mu \mathbb{S}^{d-1} \quad (13)$$

and

$$\text{Log}_\mu(\mathbf{u}) = \theta \frac{(I_d - \mu\mu^\top)(\mathbf{u} - \mu)}{\|(I_d - \mu\mu^\top)(\mathbf{u} - \mu)\|}, \quad \mathbf{u} \in \mathbb{S}^{d-1} \quad (14)$$

where $\theta = \arccos(\mathbf{u}^\top \mu)$ and $I_d \in \mathbb{R}^{d \times d}$ is the identity matrix.¹

**Thank you for your
attention!**
